TEXORA · LECTURE SHEET

Double Knit Structures: Rib and Interlock

Two needle beds double the structural possibilities. Rib alternates face and reverse wales across the width and gains balance and widthways recovery; interlock intermeshes two ribs so that both faces look like jersey and the fabric gains stability and substance. This sheet treats them as geometry — gating, needle arrangement, and the doubled yarn bookkeeping of the interlock course.

TE-3205 · FABRIC STRUCTURE AND DESIGN · TOPIC 11 · LEVEL 3 TERM 2

1. Two beds and gating

Double-bed machines carry a **cylinder** set of needles and a **dial** set at right angles to it. The relative alignment of the two sets is the **gating**: in **rib gating** the dial needles stand **between** the cylinder needles (offset by half a needle pitch), so a dial needle can draw its loop down between two cylinder wales; in **interlock gating** the dial needles stand **exactly opposite** the cylinder needles, and the two sets are worked alternately so they never collide — the gating is the first fact to state about any double-knit structure, because the structures are impossible in each other's gating.

2. Rib structure

In rib, consecutive wales are drawn to opposite faces — the notation alternates along the course (Topic 9, Figure 1) — written **face × reverse**: 1×1 rib alternates singly; 2×2 rib works two face wales then two reverse wales. Structural consequences:

- **Balance:** the face and reverse tendencies oppose within the same course, so rib does not curl — the bands and borders of garments are rib for this reason.
- **Width behaviour:** in the relaxed state the reverse wales tuck in behind the face wales, so the fabric contracts strongly across the width and extends and recovers across the width far beyond jersey — the rib fabric is considerably narrower off the machine than its needle count suggests, and WPC must always be read in the stated (relaxed or extended) condition.
- **Thickness and cover:** two beds of loops in the same courses make rib roughly twice the jersey thickness at the same count, warmer, and more opaque.

- **Unravelling:** only from the last-knitted end (the reverse loops lock the earlier courses), unlike jersey.
- **Derivatives: half cardigan** (tuck courses on one bed) and **full cardigan** (tuck on both) widen and soften the rib; **Milano rib** alternates a rib course with two jersey courses (one on each bed) for a firm, woven-like fabric.

3. Interlock structure

Interlock is **two 1×1 ribs intermeshed**: the wales of the second rib sit directly behind the wales of the first, so that **both faces of the cloth show only face (V-leg) wales** — smooth like jersey on both sides. The price is in the needle arrangement and the course accounting:

- Each bed carries **two sets of needles, long and short, alternating**.
- **Feeder 1** knits rib course A: long cylinder needles with short dial needles. **Feeder 2** knits rib course B: short cylinder with long dial — the opposite selection, its wales falling behind course A's.
- **One interlock fabric course = two feeder courses.** Any production or weight calculation that forgets this is wrong by a factor of two in the course direction.

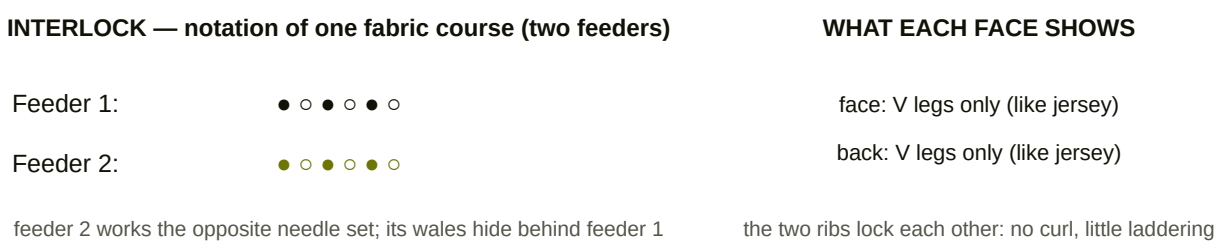


Figure 1 — Interlock in the course's dot notation (as TE-3101, Topic 2). Each feeder alone would make a 1×1 rib; intermeshed with opposite needle selection, the reverse wales of each rib are covered by the face wales of the other.

4. Worked example — interlock weight (the doubled course)

GSM OF INTERLOCK — COUNT BOTH FEEDER COURSES

Interlock, relaxed: WPC = 11 per cm, fabric CPC = 14 per cm; each rib course is knitted at stitch length 2.9 mm; yarn 19.7 tex.

Yarn per fabric course position = 2 feeder courses × 2.9 mm = 5.8 mm of yarn per wale.

GSM = WPC × CPC × (l_1 + l_2) × tex ÷ 100 = 11 × 14 × 5.8 × 19.7 ÷ 100 = 154 × 5.8 × 19.7 ÷ 100 = 17,596 ÷ 100 ≈ **176 g/m²**.

Tightness per rib course: K = √19.7 ÷ 2.9 = 4.44 ÷ 2.9 = **1.53** — each rib is knitted firmly, and the fabric carries two of them: compare the single jersey of Topic 10 at 132 g/m² from a *coarser*

> geometry. Forgetting the second feeder course would "predict" 88 g/m² — half the truth, and the standard interlock error.

Production follows the same doubling: at equal feeder count and machine speed, interlock produces about **half the length per hour** of jersey (each fabric course consumes two feeder revolutions of work), a fact TE-3101 quantified and buyers of interlock polo fabric pay for.

5. The double-knit family compared

PROPERTY	SINGLE JERSEY	1×1 RIB	INTERLOCK
Beds / gating	1 bed	2 beds, rib gating	2 beds, interlock gating, long+short needles
Faces	Different (V / arcs)	Alternating wales both sides	Both faces V wales (jersey-like)
Curl	Yes	No	No
Greatest recovery	Widthwise, moderate	Widthwise, very high	Moderate both ways
Relative weight (same count)	1 (reference)	≈1.5–2	≈2 (two courses per fabric course)
Ladders	Readily, both ends	Last end only	With difficulty
Typical use	T-shirts, dresses	Cuffs, collars, waistbands, rib bodies	Polos, quality underwear, sportswear

Glossary

Gating — alignment of dial needles relative to cylinder needles (rib: offset half-pitch; interlock: opposite). **Rib** — structure alternating face/reverse wales by course position. **Interlock** — two ribs intermeshed with opposite needle selection at alternate feeders. **Long/short needles** — the two butt-length sets enabling alternate-feeder selection. **Cardigan (half/full)** — rib with tuck courses on one/both beds.

Key points

• Gating decides the family: rib gating = dial needles between cylinder needles; interlock gating = opposite, with long/short selection.
• Rib = balance + widthways recovery + contraction; its WPC is meaningless without the fabric's condition.
• Interlock = two ribs, one fabric course = two feeder courses (GSM worked: ≈176 g/m², K per rib course 1.53) — the factor of two is the topic's main trap.

Practice

1. An interlock sample gives WPC 12, fabric CPC 15, stitch length 2.8 mm per feeder course, yarn 16.4 tex. Compute its GSM.
2. Why can a rib machine not knit interlock merely by changing the cam settings?
3. A cuff fabric must extend to twice its relaxed width and recover, and must not curl. Choose between jersey, rib and interlock and justify by two properties.
4. State the needle arrangement fault that turns intended interlock into rib, and the fabric evidence for it.

Answer key

1. GSM = 12 × 15 × (2.8 + 2.8) × 16.4 ÷ 100 = 180 × 5.6 × 16.4 ÷ 100 = 16,531 ÷ 100 ≈ **165 g/m²**.
2. Interlock needs interlock **gating** (dial needles opposite cylinder needles) and **two needle lengths per bed** with alternate-feeder selection. A rib-gated machine's needles are offset by half a pitch and are of one set; no cam change can move the needles or create the second set.
3. **1×1 rib**: its relaxed contraction stores the extension (reverse wales hidden behind face wales unfold across the width) with strong recovery, and its opposed facing makes it balanced — no curl. Jersey lacks the stored contraction; interlock's locked double structure deliberately restricts exactly this widthways give.
4. If both feeders knit on the **same** needle set (selection fault — e.g. long/short arrangement disturbed), the second feeder course knits into the same wales as an ordinary rib course: the fabric is rib, evidenced by alternating face/reverse wales visible on both faces instead of V wales everywhere.